

BIM and Safety

Building Information Modeling takes traditional 3D models that have been around for ages and adds data to them. BIM is an intelligent, 3D model-based process that provides architecture, engineering, and construction (AEC) professionals the insights and tools to efficiently plan, design, construct, and manage buildings and infrastructure. The addition of data about construction materials, costs, hazards, and building lifecycles improves collaboration and reduces errors (Autodesk University 2017).

BIM adds to safety because it makes sharing information and collaboration easier. These shared models link the trades into one model. This can help detect clashes. Software like Navisworks can link different file types such as DWG, RVT, etc. from different programs like AutoCAD, Civil3D and Revit. This can allow different models and trades to view clashes beforehand and boost collaboration (Zhang 2013). BIM models can simulate time making progress tracking and maintenance scheduling easier in 4D models (Bimsafe NZ 2024).

Hazard points, traffic areas, near misses, can all be added into the model and reassessed to make a safer job site. Models are easier to understand than written instructions. This can help eliminate language barriers and disconnect between subs/trades through visual aids (Autodesk University 2017). BIM is more than a visual aid. Object data helps improve safety. BIM can turn inspections and near misses into feedback. During a project inspection, hazardous areas or mistakes that have been made can be added to the data. Near misses can be logged as well. These updates make the model safer and in turn make the plan safer. These updates can be discussed to workers in pre-task meetings (CPWR/CDC 2016).

The main point I found was that BIM makes construction safer by planning for hazards on the front end, during the project, and maintaining its use after construction on the maintenance side.

Construction Procedures

1. Plan for safety within BIM Models - Decide what safety goals, model views, responsible parties, and when this information will be reviewed and updated. Set an outline for the safety regulations for the project.
2. Model the areas that are high risk - Temporary work, scaffolds, openings, edges, crane movement paths, access routes. Pinpoint these areas that are high risk and prioritize their safety
3. Review High Risk Work that is required on the job - once the high risk areas are ironed out, now focus on 4D models progress tracking of high risk work ex. Roofing, steel erection, concrete, edges, major lifts

4. Use the model in pre task planning and inspection - Bring the model to weekly safety meetings, daily planning, and site walks. BIM is only good when used correctly and often
5. Handover safety information to train users - Transfer maintenance and safety data to owners/FMs and make sure they can continue proper usage. (Challenges and Benefits 2025)

Sources

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Challenges and Benefits of Using BIM T Challenges and Benefits of Using BIM Technologies to Improve Construction Safety: An Exploratory Case Study Abdulaziz Alotaibi

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